

Source-native stability-label uncertainty in crystal discovery benchmarks

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Abstract

Machine-learning crystal discovery commonly evaluates models using binary stability labels derived from density-functional-theory energies above hull, yet public databases differ in workflows and hull references. We audit this source dependence on strict Materials Project–Alexandria structure matches. Of 43,984 Alexandria rows carrying Materials Project identifiers, 43,139 strict matches are retained; 5,060 have discordant source-native binary labels (11.7%; chemical-system cluster-bootstrap 95% interval, 11.3–12.1%). Discordance is chemically nonuniform and enriched where cross-source hull energies differ enough to create threshold crossings. In the clipped public representation, near-hull flags capture discordance mechanically and therefore quantify reporting burden rather than physical uncertainty. A fixed CHGNet diagnostic on 5,000 structures shows MP-minus-Alexandria precision@300/500 shifts of 4.3/3.6 percentage points. These results support source-aware benchmark reports that disclose source-conflict, near-threshold and conflict-excluded sensitivity alongside binary stability endpoints.

Keywords: crystal stability, density-functional theory, materials databases, machine learning, benchmark uncertainty

1 Introduction

Machine-learning-guided crystal discovery has moved from early property-prediction workflows to ranking and generating materials at the scale of large public databases [1–6]. These workflows require a stability label that can be applied consistently across

047 training, validation and prospective discovery, whether the model is a graph neural
048 network, a universal interatomic potential, a composition-level thermodynamic clas-
049 sifier or a generative design system [7–15]. In practice, this label is usually derived
050 from the energy above the convex hull, with structures at or below the hull treated as
051 stable and structures above the hull treated as unstable. The convention is attractive
052 because it converts a continuous thermodynamic quantity into a simple benchmark
053 endpoint [13, 16].

054 This benchmark framing is part of a broader shift toward standardized, repro-
055 ducible evaluation of materials AI methods. Toolkits and benchmarks such as
056 matminer, Matbench and JARVIS-Leaderboard have emphasized reproducible and
057 community-comparable evaluation across materials-design tasks [17–19]. Matbench
058 Discovery sharpens this perspective for crystal stability prediction by treating
059 machine-learning models as pre-screeners for DFT-based stable-materials discovery
060 and evaluating them against DFT reference convex hulls rather than model-defined
061 hulls [13]. Because binary DFT stability labels are also used in composition-based
062 stability prediction and generative inorganic-materials design, source-dependent label
063 uncertainty is not only a leaderboard issue but also affects how predicted or generated
064 candidates are interpreted [6, 15, 20–22].

065 The same convention can also make benchmark labels appear more absolute than
066 the underlying calculations justify. Public DFT databases and data-exchange infras-
067 tructures such as Materials Project, Alexandria, OQMD, JARVIS, AFLOW, NOMAD,
068 Materials Cloud and OPTIMADE are built with different workflows, pseudopo-
069 tentials, correction schemes, magnetic initializations and hull-reference conventions
070 [23–34]. These differences need not be large to matter near a hard binary threshold:
071 DFT implementations can be highly reproducible under controlled settings, yet choices
072 in exchange-correlation functional, projector augmented-wave datasets and correction
073 schemes can shift formation and hull energies by amounts that matter near zero [35–
074 44]. An energy displacement of only a few to tens of meV per atom can change a label
075 if it crosses zero, while the same displacement is irrelevant for a structure far from
076 the threshold on the same side in both sources [16, 45, 46].

077 This distinction is important for interpreting recent reports of high cross-source
078 disagreement on selected discovery subsets. Such observations could indicate a broad
079 failure of DFT labels as machine-learning ground truth, but they could also reflect
080 the source pair, the candidate-selection rule, and the fact that selection often
081 concentrates candidates near the stability boundary. A clean benchmark diagnos-
082 tic therefore requires separating exact-match source disagreement from selection-
083 conditioned probes, and separating empirical evidence from uncertainty flags that are
084 mathematically coupled to the binary label definition.

085 Here we treat this as a source-native benchmark-reporting problem rather than a
086 referendum on DFT labels. By source-native, we mean that each database is used as
087 encountered by a benchmark user: with its own public energy-above-hull value, stabil-
088 ity threshold and hull-reference convention. We separate three empirical questions and
089 one reporting implication that are easily conflated. First, how large is MP–Alexandria
090 binary stability-label discordance on a strict shared-structure denominator? Second,

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is that discordance associated with the real-valued cross-source hull-energy difference rather than only with a boundary flag coupled to the label definition? Third, how large is the benchmark readout sensitivity induced by changing only the label source? The reporting implication is how public benchmarks should disclose source-sensitive and near-threshold labels without treating the clipped public representation as mechanism-resolved evidence. We show that full-denominator discordance is modest but non-negligible, that the estimate remains stable under cluster bootstrap uncertainty, that discordance is enriched at intermediate-to-large cross-source energy differences and in some chemical subsets, and that label-source choice imposes both an oracle benchmark ceiling and measurable fixed-ranking precision shifts. Near-hull flags in the available public-label representation are best interpreted as reporting-burden diagnostics. Selection-conditioned WBM–Alexandria probes are retained only as supplementary diagnostics, not as source-wide claims.

2 Results

2.1 Full MP–Alexandria MP-identifier exact matching defines the comparison denominator

We first constructed a denominator in which both sources refer to the same atomic structure rather than merely the same reduced formula. The local Alexandria v20 snapshot contains 675,204 train+val rows, of which 43,984 unique rows carry Materials Project identifiers after duplicate MP identifiers are removed. We queried the corresponding Materials Project records for the full MP-ID overlap; 43,169 records were successfully retrieved, while 815 identifiers were not returned by the Materials Project query used for this run. Candidate pairs were prefiltered by reduced formula and compared using `pymatgen StructureMatcher` [47] with `ltol=0.2`, `stol=0.3`, `angle_tol=5`, `primitive_cell=True`, `scale=True` and `attempt_supercell=True`. Of the queried records, 43,139 passed strict structure matching and 30 failed strict matching. These 43,139 pairs define the primary MP–Alexandria source-native stability-label denominator (Fig. 1; Table 1).

The previously used 287-structure exact-match audit is therefore demoted to a scored diagnostic subset for analyses requiring frozen model scores and legacy selection-conditioned probes. The full denominator should still be read as an MP-identifier overlap, not as a random intersection of independently sampled discovery sets or as a census of all Alexandria structures. We also attempted third-source triangulation on the diagnostic workflow. OQMD exact-structure coverage was only $n=4$, far below the scale required for a statistically meaningful third-source measurement. We therefore report OQMD as an explicit coverage boundary rather than a validation source.

2.2 Overall and chemistry-stratified source-native discordance

Across the full MP–Alexandria exact-match denominator, binary stability-label discordance is 11.7% (5,060/43,139; exact binomial 95% confidence interval, 11.4–12.0%).

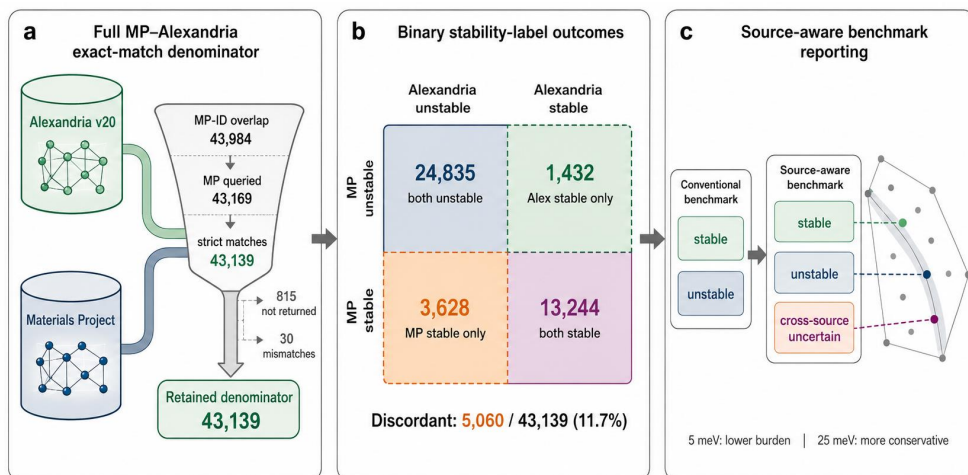


Fig. 1 Full-denominator exact matching exposes a modest but non-negligible source-discordance baseline. **a**, The source-comparison denominator is constructed from Alexandria v20 rows carrying Materials Project identifiers and retains 43,139 strict MP-Alexandria structure matches. **b**, Source-native binary stability labels agree for most matched structures, but 5,060/43,139 are discordant. **c**, The benchmark-reporting implication is to disclose source-sensitive or near-threshold label burden alongside the conventional binary endpoint; in the public-label setting studied here this is a reporting diagnostic, not a replacement leaderboard.

Table 1 Full MP-Alexandria denominator construction audit. The 43,139 matched structures are a unique MP-identifier exact-match denominator, not a random intersection of independently sampled discovery sets.

Step	Count	Role in analysis
Alexandria v20 entries	675,204	Full local train+val rows
Unique Alexandria entries with MP identifiers	43,984	Candidate universe for MP-ID joins
Materials Project records successfully queried	43,169	Queryable MP-ID overlap
MP-ID query gap	815	Not returned by the MP query used here
Strict MP-Alexandria structure matches	43,139	Primary source-native denominator
Structure mismatches among queried records	30	Excluded from denominator
Scored diagnostic subset	287	ML and legacy probe diagnostics
OQMD exact matches in diagnostic workflow	4	Coverage boundary, not a third-source estimate

Because database entries are chemically correlated, the binomial interval is descriptive rather than the only uncertainty summary. Cluster bootstrapping by chemical system gives a 95% interval of 11.3–12.1%, and bootstrapping by reduced formula gives 11.4–12.1%; both support the same modest but non-negligible source-pair baseline. The confusion matrix shows that most structures retain the same binary label in both sources: 24,835 are unstable in both sources and 13,244 are stable in both sources (Fig. 1b). Directionally, discordance is asymmetric: 3,628 pairs are MP-stable/Alexandria-unstable, whereas 1,432 pairs are MP-unstable/Alexandria-stable.

This directionality is consistent with source-specific correction schemes and hull-reference conventions [23, 24]; we do not attribute it further because the study intentionally uses each source’s native public labels. Thus the full-denominator source-pair baseline is well below the high-discordance regime suggested by selected-probe observations, but it is not negligible for benchmark protocols that treat binary labels as source-invariant. The earlier 287-structure scored audit gave a similar estimate (31/287, 10.8%) and is used below only where full-denominator model scores or legacy probe diagnostics are unavailable.

The global rate also masks chemical variation (Fig. 2; Table 2). Discordance is 12.6% for binary systems, 12.7% for ternaries and 8.2% for quaternary-or-higher systems; the unary subset is small but has 15.2% discordance. Element-family stratification shows that halogen-containing systems have higher discordance than non-halogen systems (15.9% versus 11.0%), whereas oxygen-containing systems have lower discordance than oxygen-free systems (7.3% versus 13.3%). Lanthanide-containing systems are modestly elevated (13.7% versus 10.7% without lanthanides), while transition-metal-containing systems are slightly lower than the complement (11.1% versus 13.7%). These contrasts are descriptive rather than mechanistic: they identify where source-native labels differ in this overlap, but they do not isolate whether the difference comes from formation energies, corrections, magnetic treatments or hull-reference sets.

Direction-specific energy-difference distributions further show that discordant cases are not simply random draws from the concordant population. Concordant stable pairs have $|\Delta e| = 0$ in this clipped public representation, while concordant unstable pairs have median $|\Delta e| = 2.6$ meV atom⁻¹. The two discordance directions have substantially larger median separations: 10.0 meV atom⁻¹ for MP-stable/Alexandria-unstable pairs and 9.0 meV atom⁻¹ for MP-unstable/Alexandria-stable pairs. The 815 MP-ID query-gap records are also archived as a bias audit rather than silently dropped: 327 are Alexandria-stable and 488 are Alexandria-unstable, span 703 chemical systems, have median 12 atoms per cell and include 539 transition-metal-containing, 491 lanthanide-containing, 259 oxygen-containing and 127 halogen-containing entries.

2.3 Cross-source hull-energy differences and reporting burden

We next asked whether binary label discordance is associated with the underlying real-valued disagreement between the two DFT sources, rather than with a near-hull flag that is mathematically coupled to the label definition. For each matched prototype we computed $\Delta e = e_{\text{hull}}^{\text{MP}} - e_{\text{hull}}^{\text{Alex}}$ and stratified the pairwise discordance rate by $|\Delta e|$. Discordance is lower when the two sources agree to within 10 meV atom⁻¹ (2,559/34,442, 7.4%; exact binomial 95% CI, 7.2–7.7%), but rises to 28.8% (2,390/8,287; 95% CI, 27.9–29.8%) for structures with $10 \leq |\Delta e| < 100$ meV atom⁻¹. The ≥ 100 meV atom⁻¹ tail remains elevated (111/410, 27.1%; 95% CI, 22.8–31.7%), confirming that cross-source energy separation is associated with threshold crossing but is not sufficient for it. Thus the empirical signal is best described as a cross-source energy-difference tail that becomes consequential when it straddles the stability threshold.

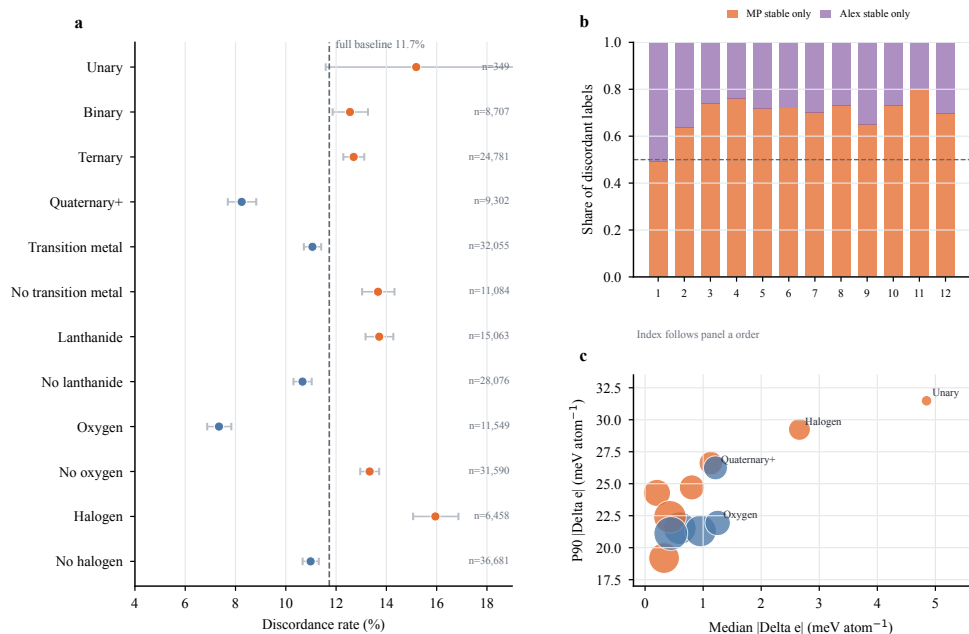


Fig. 2 Source-native discordance is chemically nonuniform but not confined to a single chemistry class. **a**, Discordance rates for element-count and element-family subsets. Points show subset rates, horizontal bars show exact-binomial 95% intervals and the dashed line marks the 11.7% full-denominator baseline. **b**, Directional composition of discordant labels in the same subset order as panel a. **c**, Median and 90th-percentile absolute cross-source hull-energy differences for the same subsets; marker area scales with subset size. These panels are descriptive diagnostics, not mechanism-resolved attribution to a specific DFT workflow difference.

The either-source near-hull flag is not independent evidence for localization in the available public-label representation. In the matched public tables used here, stable or on-hull entries are represented at exactly $e_{\text{hull}} = 0$ for both sources (16,872/16,872 MP-stable entries and 14,676/14,676 Alexandria-stable entries), while unstable entries are positive. Under this clipped representation, any binary discordance necessarily has one source at zero and therefore mechanically triggers an either-source near-hull flag at any positive threshold. We therefore interpret the threshold sweep as a benchmark-reporting burden analysis, not as causal evidence that the flag explains discordance or as a refined uncertainty-localization method.

The burden changes substantially with the cutoff. A 5 meV atom⁻¹ either-source flag captures all 5,060 observed discordant pairs while flagging 21,354/43,139 structures; a 25 meV atom⁻¹ flag also captures all 5,060 discordant pairs but flags 28,985/43,139 structures (Fig. 3c). In both cases, the outside-flagged set has zero observed discordance, but this follows from the clipped stable labels rather than from a newly localized physical boundary. The practical recommendation is therefore disclosure rather than a universal three-state rule: public benchmark reports should state how many labels are source-sensitive or near-threshold under the representation

Table 2 Chemistry-stratified MP–Alexandria source-native discordance. Rates are descriptive full-denominator summaries; overlapping element-family rows are not independent categories.

Subset	<i>n</i>	Discordant	Rate	Directional counts
Unary	349	53	15.2%	26 MP-only stable; 27 Alex-only stable
Binary	8,707	1,093	12.6%	696 MP-only stable; 397 Alex-only stable
Ternary	24,781	3,147	12.7%	2,323 MP-only stable; 824 Alex-only stable
Quaternary+	9,302	767	8.2%	583 MP-only stable; 184 Alex-only stable
Transition metal present	32,055	3,545	11.1%	2,539 MP-only stable; 1,006 Alex-only stable
Lanthanide present	15,063	2,066	13.7%	1,445 MP-only stable; 621 Alex-only stable
Oxygen present	11,549	848	7.3%	551 MP-only stable; 297 Alex-only stable
Halogen present	6,458	1,030	15.9%	821 MP-only stable; 209 Alex-only stable

available to users, and stronger three-state annotations should be based on unclipped continuous energies or common-hull reconstructions where possible.

2.4 Label-source choice shifts oracle and CHGNet benchmark readouts

We next asked whether source-native label choice has measurable benchmark consequences, rather than only producing an audit table of disagreements. A perfect MP-label oracle, evaluated against MP labels, has F1=1.000 by definition. Evaluating the same MP labels against Alexandria labels reduces F1 to 0.8396 and accuracy to 0.8827 (Fig. 4a). Equivalently, excluding all source-conflict entries retains 38,079/43,139 structures (88.27%) while removing the 5,060 label-sensitive cases. This is an upper-bound label-transfer diagnostic, not an ML model result, but it quantifies the metric ceiling imposed by source-native label non-interchangeability.

To test whether the same issue appears under a real model-facing ranking, we scored a deterministic 5,000-structure subset of the full denominator with CHGNet [48]. The 5,000-structure sample is representative of the full denominator at the level needed for this diagnostic: its MP-stable rate is 39.3%, Alexandria-stable rate is 34.6% and MP–Alexandria discordance is 11.2%, close to the corresponding full-denominator values of 39.1%, 34.0% and 11.7%. We used the frozen CHGNet formation-energy proxy only as a model-facing ranking, not as a Matbench-Discovery-style reconstructed-hull predictor or as a model leaderboard. A sign-sanity check confirms this scope: raw CHGNet formation-energy proxy AUROC is 0.546 against MP labels and 0.534 against Alexandria labels, while the negative-proxy direction used for the frozen ranking has AUROC 0.454 and 0.466, respectively. These weak threshold-free values show that the diagnostic should not be interpreted as CHGNet stability performance. The diagnostic deliberately uses a weak fixed ranking because the estimand is label-source sensitivity under identical candidates and scores, not model quality.

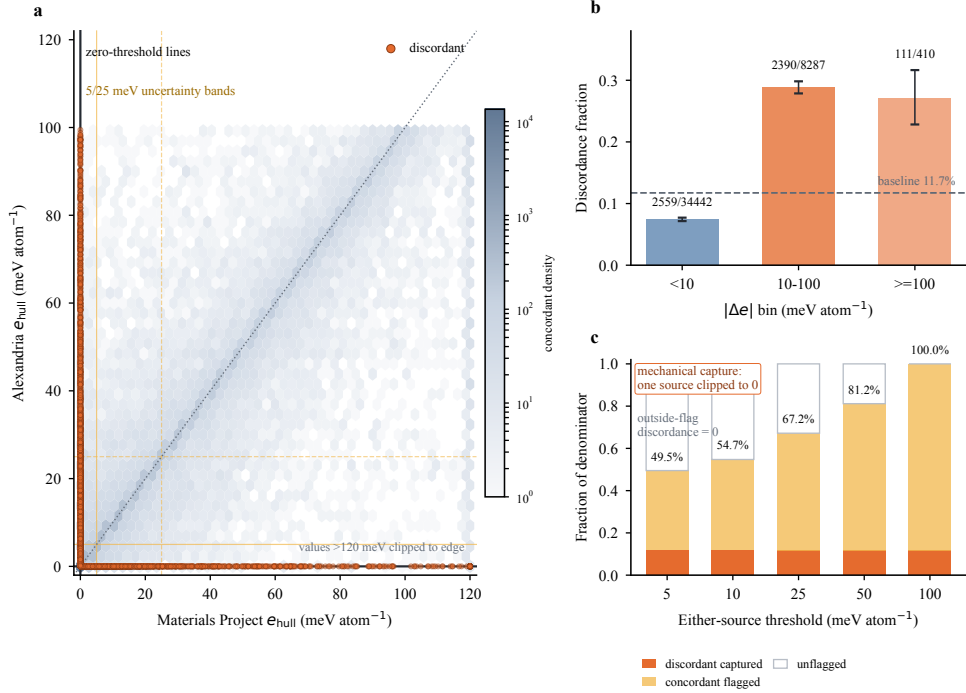


Fig. 3 Cross-source hull-energy differences identify where label discordance is enriched and calibrate the cost of uncertainty reporting. **a**, MP and Alexandria source-native hull energies for 43,139 strict matches. The density layer shows concordant structures and orange points mark discordant threshold crossings; values above 120 meV atom $^{-1}$ are clipped to the panel edge for readability. **b**, Discordance fraction by absolute cross-source hull-energy difference. Bars show exact-binomial 95% confidence intervals and the dashed line marks the 11.7% full-denominator baseline. **c**, In the clipped public-label representation, either-source near-hull flags mechanically capture all observed discordant pairs at positive thresholds; the panel therefore reports annotation burden, with the flagged fraction rising from 49.5% at 5 meV atom $^{-1}$ to 67.2% at 25 meV atom $^{-1}$.

Under the same fixed CHGNet ranking, however, the evaluation readout changes when only the label source changes. Precision@300 is 0.333 against MP labels and 0.290 against Alexandria labels, a 4.3 percentage-point MP-minus-Alexandria shift with bootstrap 95% interval 0.7–8.0 percentage points. Precision@500 shifts from 0.304 to 0.268, a 3.6 percentage-point shift with bootstrap 95% interval 1.2–6.2 percentage points. The K=300 top set contains 77 structures stable in both sources, 23 MP-only stable structures, 10 Alexandria-only stable structures and 190 structures unstable in both sources, giving MP stable count 100 versus Alexandria stable count 87 under identical candidates and scores (Fig. 4f). Thus label-source choice alone imposes both an oracle benchmark ceiling and measurable fixed-ranking precision shifts.

Secondary scored-subset and WBM–Alexandria diagnostics are reported in Supplementary Note 1 to document selection-conditioning effects; they are not used as source-wide estimates or as evidence for model behavior on the 43,139-structure denominator.

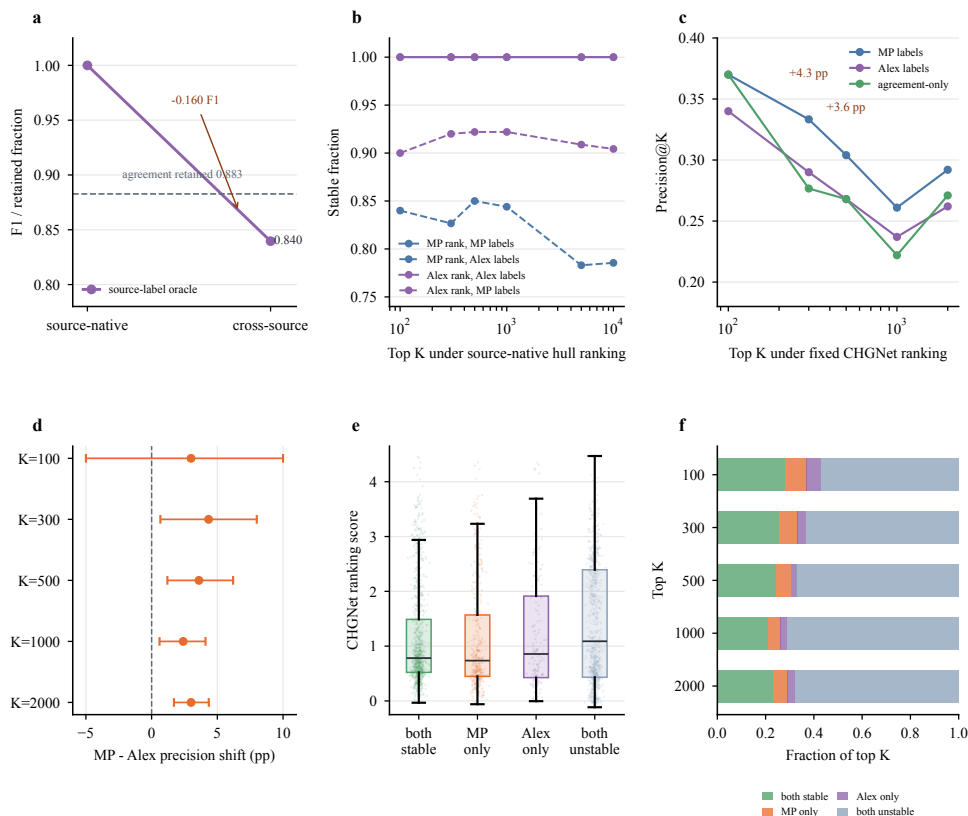


Fig. 4 Source-native label choice changes benchmark readouts even when predictions are fixed. **a**, A perfect source-label oracle loses 0.160 F1 under cross-source evaluation; the dashed line shows that source-agreement-only reporting retains 88.27% of the denominator. **b**, Source-native hull-energy rankings show different top- K stable fractions when evaluated with the other source’s labels. **c**, Under a fixed CHGNet formation-energy-proxy ranking on 5,000 deterministic structures, precision@ K differs when the evaluation labels are switched from MP to Alexandria. **d**, Bootstrap 95% intervals for the MP-minus-Alexandria precision shift. **e**, CHGNet ranking-score distributions stratified by source-label class. **f**, Top- K source-label composition under the same CHGNet ranking. The CHGNet diagnostic is a model-facing label-source sensitivity check, not a composition-aware stability leaderboard.

3 Discussion

This study separates three empirical questions and one reporting implication that are easily conflated in ML-driven crystal-discovery benchmarks. The first empirical question is the source-pair baseline within a strict MP-identifier denominator: across 43,139 MP–Alexandria exact matches retained from 43,984 Alexandria rows carrying MP identifiers, binary stability-label discordance is 11.7%, with cluster-bootstrap uncertainty consistent with the exact-binomial summary. The second is the source-energy and chemistry structure of that discordance: it is enriched when cross-source hull-energy differences are large enough to create threshold crossings, and its rate

varies across chemical subsets. The third is whether label-source choice changes benchmark readouts under fixed predictions: it does, both for a perfect source-label oracle and for a 5,000-structure CHGNet ranking diagnostic. The reporting implication is how benchmark protocols should disclose this uncertainty to users rather than hiding it inside a binary endpoint.

The physical and statistical interpretation is straightforward, but it should not be overread. The stability label is a hard threshold applied to a real-valued quantity. Different DFT workflows can shift that quantity through choices in pseudopotentials, correction schemes, magnetic states, convergence settings and hull references. Those shifts are often small on the scale of formation energies, but they are decisive when a structure sits close enough to the hull that zero lies between the two source estimates. This is why a label can be unstable across sources even when neither source is obviously erroneous. The result is therefore not evidence that DFT itself is inconsistent for these structures. It is evidence that public source-native DFT-derived stability labels are not always interchangeable.

The appropriate benchmark response is therefore not to abandon binary stability labels. Binary labels remain discriminative for many structures and are useful for leaderboard-scale comparisons, but they should not be presented as source-invariant near the hull. In the clipped public-label setting studied here, however, the either-source near-hull flag is too mechanically coupled to the label definition to serve as a refined uncertainty-localization method: a 5 meV atom⁻¹ flag captures all observed discordant pairs while flagging 21,354/43,139 structures, and a 25 meV atom⁻¹ flag captures the same discordant set while flagging 28,985/43,139. The practical recommendation is correspondingly narrower. Public benchmark reports should disclose how many labels are source-sensitive or near-threshold under the representation available to users. A stronger stable/unstable/cross-source-uncertain annotation would require unclipped continuous energies, a common thermodynamic hull, or a protocol that can distinguish public-table clipping from physically meaningful proximity to the hull.

The benchmark-impact diagnostics make the same point in evaluation units. A perfect MP-source labeler transfers to Alexandria labels with F1=0.8396 rather than 1.000, and the source-agreement-only denominator retains 88.27% of the full denominator. In the CHGNet sensitivity check, the same ranked 5,000 structures yield precision@300/500 shifts of 4.3 and 3.6 percentage points depending only on whether MP or Alexandria labels are used for evaluation. These values are not large enough to imply a universal model-ranking reversal, and the CHGNet formation-energy proxy is deliberately not promoted as a composition-aware stability score. Their importance is more basic: label-source choice creates a measurable uncertainty band in top-K benchmark readouts even before model architecture comparisons are considered.

This source-native analysis also conflates mechanisms that would need additional data to separate. The observed label discordance may reflect differences in DFT total energies or formation energies, compatibility corrections, phase-diagram reference sets, magnetic initialization or ordering, oxidation-state assumptions, database versioning and API availability. The present public-safe tables do not contain the full unclipped formation-energy records, competing phase diagrams and workflow metadata needed to decompose those channels quantitatively. We therefore interpret the result as a

user-facing public-label reproducibility audit, not as a mechanism-resolved comparison of underlying DFT calculations. Recomputing matched entries on a common hull, or comparing source-native labels with common-hull labels built from unclipped formation energies, is the natural next step.

Our analysis is complementary to recent work on model-side and data-side reliability of universal MLIPs. Systematic softening studies diagnose how pretrained potentials can distort high-energy or out-of-distribution regions of the potential-energy surface, while high-fidelity and off-equilibrium datasets such as MP-ALOE aim to improve the training coverage that controls those errors [49, 50]. Subsequent foundation-MLIP work has also begun to report performance directly on Matbench Discovery, indicating that crystal-stability discovery benchmarks are becoming a shared evaluation language [51]. By contrast, the present work holds the public source-native labels fixed and asks whether the binary stability endpoint changes across DFT databases for the same matched structure. These are distinct uncertainty channels and should be reported separately.

These limitations define the intended scope rather than invalidating the result. This study is not a universal discordance estimate for all DFT databases or all discovery pipelines. It is a source-native public-label reproducibility audit on a stringent MP-identifier exact-match denominator. The denominator likely over-represents prototypes computed or curated by both sources; independently sampled discovery sets may exhibit different discordance behavior. The primary analysis uses two PBE-family public sources; OQMD exact-structure coverage was too small for completed three-source triangulation. The analysis uses each source’s own hull reference and does not standardize all entries onto a common thermodynamic phase diagram. The CHGNet analysis is a model-facing label-source sensitivity check on a deterministic 5,000-structure subset, not a full-denominator model audit or a Matbench-Discovery-style composition-aware leaderboard. Finally, uncertainty flags in the clipped public-label representation are benchmark-reporting diagnostics, not claims of physical instability, and thermodynamic stability is not synthesizability. Future work should combine full-denominator label-source audits with common-hull reconstructions, unclipped formation-energy comparisons and full-denominator scoring of newer foundation MLIPs such as GRACE or cross-domain models, but those extensions are outside the scope of the present source-label reproducibility study [51, 52].

4 Methods

4.1 Data sources

Materials Project and Alexandria entries were compared using source-provided energies above hull and binary stability labels derived from those energies. Alexandria labels came from the local alex-mp v20 snapshot (`alex_mp_20.zip`), which contains 675,204 train+val rows, including 43,984 unique rows whose `material_id` carries a Materials Project identifier after duplicate MP identifiers are removed. Materials Project records were queried on 2026-05-20 for all 43,984 unique MP identifiers using the Materials Project API `MPRester.materials.summary.search` endpoint with fields `material_id`, `formula_pretty`, `energy_above_hull` and `structure`. The

507 query returned 43,169 records; 815 identifiers were recorded as missing MP records for
508 this run because they were not returned by that summary endpoint. The missing-ID
509 audit contains 327 Alexandria-stable and 488 Alexandria-unstable rows distributed
510 across 703 chemical systems, with median 12 atoms per cell; this audit table is archived
511 with the full-denominator outputs. Among the queried records, 43,139 passed strict
512 structure matching and 30 failed strict matching. Accordingly, the primary analy-
513 sis estimates source-native MP–Alexandria label discordance on the full queryable
514 MP-identifier exact-match denominator. It should not be interpreted as a random
515 intersection of independently discovered structures or as a census of all Alexandria
516 rows without MP identifiers.

517 The earlier 287-structure denominator is retained only as a scored diagnostic sub-
518 set. It used a predeclared set of 300 Alexandria MP-identifier candidates selected
519 deterministically before outcome analysis by concatenating the Alexandria validation
520 and training tables, retaining rows with `material_id` beginning with `mp-`, dropping
521 duplicate identifiers, sorting by `material_id`, and taking the first 300 records under
522 the frozen diagnostic-subset limit. No discordance, model-score or uncertainty-flag
523 outcome was used in selecting this subset; 287 candidates with available MP records
524 passed strict structure matching. OQMD was queried on 2026-05-20 for an exact-
525 structure coverage check in this diagnostic workflow; exact matches were $n=4$ and
526 were not used for the primary estimate. The archived analysis tables and manifests
527 record row counts and SHA256 hashes for derived public-safe tables.

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529 4.2 Structure matching

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531 Candidate pairs were first filtered by reduced formula. Surviving pairs were
532 compared with `pymatgen.analysis.structure_matcher.StructureMatcher`
533 [47] using `ltol=0.2`, `stol=0.3`, `angle_tol=5`, `primitive_cell=True`,
534 `scale=True` and `attempt_supercell=True`. Only pairs with
535 `match_status=strict_structure_match` were retained for the primary
536 MP–Alexandria analysis.

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538 4.3 Stability labels

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540 Binary stability was defined within each source as $e_{\text{hull}} \leq 0$ eV/atom using the values
541 reported in the public matched tables. Because the analysis uses public table values
542 rather than recomputed continuous hull distances, no additional floating-point toler-
543 ance was applied beyond the source-provided zero convention. Each source’s own hull
544 reference was retained because the purpose of the study is to measure cross-source
545 label behavior under source-native public labels. In the matched tables used for this
546 analysis, stable entries are reported at exactly $e_{\text{hull}} = 0$ for both sources, whereas
547 unstable entries have positive e_{hull} values. This public-table representation is used for
548 the boundary-flag calibration below. Retaining source-native hull references is also
549 a limitation: it measures label reproducibility as a user would encounter it, not the
550 residual disagreement after restandardizing all calculations onto a shared hull.

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4.4 Cross-source energy differences

For matched prototype i , we computed $\Delta e_i = e_{\text{hull},i}^{\text{MP}} - e_{\text{hull},i}^{\text{Alex}}$ and summarized discordance over pre-specified $|\Delta e|$ bins: 0–5, 5–10, 10–25, 25–50, 50–100 and ≥ 100 meV/atom. We also report a compressed contrast between $|\Delta e| < 10$ meV/atom and $10 \leq |\Delta e| < 100$ meV/atom because it captures the main enrichment while avoiding overinterpretation of sparse extreme-tail bins.

4.5 Chemistry-stratified diagnostics

Chemical systems were parsed from the source chemical-system strings, and element-count bins were defined as unary, binary, ternary and quaternary-or-higher. Element-family indicators were computed for transition metals, lanthanides, actinides, oxygen and halogens using standard element sets. Direction-specific summaries used four label outcomes: both stable, both unstable, MP-stable/Alexandria-unstable and MP-unstable/Alexandria-stable. Missing Materials Project records were summarized separately by Alexandria label, chemical-system count, cell size and element-family indicators to assess whether the query gap was chemically concentrated.

4.6 Boundary-flag definitions

Boundary flags were computed at 5, 10, 25, 50 and 100 meV atom⁻¹ using the either-source minimum absolute hull distance, $\min(|e_{\text{hull}}^{\text{MP}}|, |e_{\text{hull}}^{\text{Alex}}|)$. In the clipped matched public tables used here, discordant pairs necessarily have one source reported at $e_{\text{hull}} = 0$. Threshold sweeps therefore quantify the number of structures flagged and the remaining outside-flag discordance rate, not independent causal localization or a general three-state annotation rule.

4.7 Model-facing benchmark sensitivity diagnostics

The main model-facing diagnostic used CHGNet [48] to score a deterministic 5,000-structure subset of the full MP–Alexandria strict denominator. The subset was sampled with seed 20260523 from the 43,139 strict matches and scored in batches of 64 on a CUDA device using `CHGNet.load()`. The archived manifest records the model identifier, sample seed, device and local Materials Project structure-cache hash. CHGNet structure energies were converted to a formation-energy proxy using Materials Project elemental reference structures; the negative proxy was used as the frozen ranking for precision@K readouts. This score is not a composition-aware hull distance, and the diagnostic is not a Matbench-Discovery-style leaderboard.

For sign and scope control, we also computed AUROC/AUPRC for the raw formation-energy proxy, the negative proxy and a random ranking baseline against both MP and Alexandria labels. The raw proxy has AUROC 0.546 against MP labels and 0.534 against Alexandria labels; the negative proxy has AUROC 0.454 and 0.466. These values are reported only to rule out a coding-sign artifact and to show that raw CHGNet formation energy is a weak cross-composition stability proxy in this setting.

The earlier ALIGNN-FF, CHGNet and MACE-MP scores were evaluated on the 287-structure scored diagnostic subset and are retained as secondary diagnostics.

ALIGNN-FF used the local archive v12.2.2024_dft_3d_307k.zip with analysis-environment packages alignn==2026.4.2 and jarvis-tools==2026.4.2. CHGNet used chgnet==0.3.0. MACE-MP used the local Materials Project MACE checkpoint 20231210mace128L0_energy_epoch249model [14]. For each model, structures were ranked by frozen score and the MP-Alexandria discordance fraction was evaluated over the top 1, 5, 10, 20 and 50% of each ranking. The top decile was defined as $\lceil 0.10n \rceil = 29$ structures for $n = 287$.

4.8 Uncertainty-filtered benchmark diagnostics

Benchmark-style metrics were computed on the 287-structure scored diagnostic subset and on subsets remaining after either-source near-hull flags at 5, 10, 25 and 50 meV atom⁻¹. Models were converted to binary predictions by taking the top 20 structures under each frozen score as predicted stable. We report precision, recall, F1, balanced accuracy, top-K stable rate and supporting AUROC where both classes are present. These filtered metrics are diagnostic because the clipped public-label representation can remove all stable positives from the unflagged subset.

4.9 Probe reconciliation diagnostics

The MP-native selection-conditioned diagnostic selected all full-denominator matched structures with $e_{\text{hull}}^{\text{MP}} \leq 0$ and evaluated binary disagreement against Alexandria labels. A near-hull sensitivity diagnostic selected full-denominator structures with $e_{\text{hull}}^{\text{MP}} \leq 25$ meV atom⁻¹. The WBM-Alexandria probe was formalized from 701 phase29 candidate rows, of which 270 passed exact structure matching against Alexandria; formula-only rows were excluded from the discordance denominator. The archived formal-definition table records the WBM source, phase29 candidate-universe identifier, matching denominator and binary-label fields used for this case analysis. The strict release-rule portability check evaluated the original release rule on the 287-structure scored diagnostic subset across 100 seed-budget rows; this check was used only to assess rule portability in the archived secondary diagnostics.

4.10 Statistics

Exact binomial 95% confidence intervals are reported as descriptive intervals for row-level discordance fractions. To account for chemical correlation among database entries, the primary discordance rate was also summarized with nonparametric cluster bootstraps by chemical system and by reduced formula. Each bootstrap resampled clusters with replacement for 5,000 iterations using seed 20260522 and recomputed the row-weighted discordance rate. Full-denominator analyses use the 43,139 strict MP-Alexandria matches. CHGNet precision-shift intervals resampled the deterministic 5,000-structure scored subset with replacement for 2,000 bootstrap iterations using seed 20260524 and recomputed MP-minus-Alexandria precision@K under the same frozen ranking. Selection-fraction analyses use frozen model-score rankings on the 287-structure scored diagnostic subset. Logistic diagnostics used standardized predictors for discordance, including $|\Delta e|$, source-native hull energies, minimum absolute

hull distance and model rank percentile. Rank-percentile p values were estimated by 2,000 rank-permutation tests with the remaining predictors fixed.

Data availability

The matched summary tables, figure source data and derived public-safe analysis outputs used in this study are archived at Zenodo, DOI: <https://doi.org/10.5281/zenodo.20392665>. The archive contains the full MP–Alexandria denominator outputs, CHGNet 5,000-structure model-facing sensitivity diagnostics, scored-subset diagnostics, figure source data, manuscript figures, file list and SHA256 manifest. The public development repository is available at <https://github.com/Marchemantics/materials-stability-label-discordance>. The raw Materials Project structure cache used during denominator construction is a local API cache and is not redistributed; the public-safe matched summary tables can be regenerated from public source records using the archived scripts subject to the Materials Project API terms.

Code availability

Code used to generate the analysis tables, model-score diagnostics, full MP–Alexandria denominator and manuscript figures is archived with the Zenodo record, DOI: <https://doi.org/10.5281/zenodo.20392665>, and is also available in the public development repository at <https://github.com/Marchemantics/materials-stability-label-discordance>. The archived code includes the full-denominator construction, major-revision diagnostics, CHGNet sensitivity check, benchmark-impact analysis, figure-building scripts and integrity tests.

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Author contributions

X.W. and J.Z. contributed equally to this work. J.Z. curated the data, performed the analysis and prepared the figures. X.W., J.Z., Y.C. and Y.H. contributed to the study framing, interpretation of results and manuscript revision. Y.C. supervised the project.

Competing interests

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies

The authors used OpenAI Codex/ChatGPT only for data-analysis support, computational-check scripting and method/background research. The authors reviewed all AI-assisted analytical outputs and take full responsibility for the content.

References

- [1] Ward, L., Agrawal, A., Choudhary, A. & Wolverton, C. A general-purpose machine learning framework for predicting properties of inorganic materials. *npj Computational Materials* **2**, 16028 (2016).
- [2] Butler, K. T., Davies, D. W., Cartwright, H., Isayev, O. & Walsh, A. Machine learning for molecular and materials science. *Nature* **559**, 547–555 (2018).
- [3] Schmidt, J., Marques, M. R., Botti, S. & Marques, M. A. Recent advances and applications of machine learning in solid-state materials science. *npj computational materials* **5**, 83 (2019).
- [4] de Pablo, J. J. *et al.* New frontiers for the materials genome initiative. *npj Computational Materials* **5**, 41 (2019).
- [5] Merchant, A. *et al.* Scaling deep learning for materials discovery. *Nature* **624**, 80–85 (2023).
- [6] Zeni, C. *et al.* A generative model for inorganic materials design. *Nature* **639**, 624–632 (2025).
- [7] Xie, T. & Grossman, J. C. Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Physical review letters* **120**, 145301 (2018).
- [8] Chen, C., Ye, W., Zuo, Y., Zheng, C. & Ong, S. P. Graph networks as a universal machine learning framework for molecules and crystals. *Chemistry of Materials* **31**, 3564–3572 (2019).
- [9] Schütt, K. T., Sauceda, H. E., Kindermans, P.-J., Tkatchenko, A. & Müller, K.-R. SchNet—a deep learning architecture for molecules and materials. *The Journal of chemical physics* **148** (2018).
- [10] Chen, C. & Ong, S. P. A universal graph deep learning interatomic potential for the periodic table. *Nature Computational Science* **2**, 718–728 (2022).
- [11] Batzner, S. *et al.* E (3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature communications* **13**, 2453 (2022).

[12]	Musaelian, A. <i>et al.</i> Learning local equivariant representations for large-scale atomistic dynamics. <i>Nature Communications</i> 14 , 579 (2023).	737 738 739
[13]	Riebesell, J. <i>et al.</i> A framework to evaluate machine learning crystal stability predictions. <i>nature machine intelligence</i> 7 , 836–847 (2025).	740 741 742
[14]	Batatia, I. <i>et al.</i> A foundation model for atomistic materials chemistry. <i>The Journal of chemical physics</i> 163 (2025).	743 744 745
[15]	Zou, H. <i>et al.</i> Predicting thermodynamic stability of inorganic compounds using ensemble machine learning based on electron configuration. <i>Nature Communications</i> 16 , 203 (2025).	746 747 748
[16]	Sun, W. <i>et al.</i> The thermodynamic scale of inorganic crystalline metastability. <i>Science advances</i> 2 , e1600225 (2016).	749 750 751
[17]	Ward, L. <i>et al.</i> Matminer: An open source toolkit for materials data mining. <i>Computational Materials Science</i> 152 , 60–69 (2018).	752 753 754
[18]	Dunn, A., Wang, Q., Ganose, A., Dopp, D. & Jain, A. Benchmarking materials property prediction methods: the matbench test set and automatminer reference algorithm. <i>npj Computational Materials</i> 6 , 138 (2020).	755 756 757 758
[19]	Choudhary, K. <i>et al.</i> Jarvis-leaderboard: a large scale benchmark of materials design methods. <i>npj Computational Materials</i> 10 , 93 (2024).	759 760 761
[20]	Noh, J. <i>et al.</i> Inverse design of solid-state materials via a continuous representation. <i>Matter</i> 1 , 1370–1384 (2019).	762 763 764
[21]	Xie, T., Fu, X., Ganea, O.-E., Barzilay, R. & Jaakkola, T. S. Crystal diffusion variational autoencoder for periodic material generation. <i>International Conference on Learning Representations</i> 1–17 (2022). URL https://openreview.net/forum?id=03RLpj-tc_ .	765 766 767 768
[22]	Jiao, R. <i>et al.</i> Crystal structure prediction by joint equivariant diffusion. <i>Advances in Neural Information Processing Systems</i> 36 , 17464–17497 (2023).	769 770 771
[23]	Jain, A. <i>et al.</i> Commentary: The materials project: A materials genome approach to accelerating materials innovation. <i>APL materials</i> 1 (2013).	772 773 774
[24]	Schmidt, J. <i>et al.</i> Machine-learning-assisted determination of the global zero-temperature phase diagram of materials. <i>Advanced Materials</i> 35 , 2210788 (2023).	775 776 777
[25]	Saal, J. E., Kirklin, S., Aykol, M., Meredig, B. & Wolverton, C. Materials design and discovery with high-throughput density functional theory: the open quantum materials database (oqmd). <i>Jom</i> 65 , 1501–1509 (2013).	778 779 780 781 782

783 [26] Kirklin, S. *et al.* The open quantum materials database (oqmd): assessing the
784 accuracy of dft formation energies. *npj Computational Materials* **1**, 1–15 (2015).
785

786 [27] Choudhary, K. *et al.* The joint automated repository for various integrated sim-
787 ulations (jarvis) for data-driven materials design. *npj computational materials* **6**,
788 173 (2020).

789 [28] Curtarolo, S. *et al.* Aflow: An automatic framework for high-throughput materials
790 discovery. *Computational Materials Science* **58**, 218–226 (2012).
791

792 [29] Curtarolo, S. *et al.* Aflowlib. org: A distributed materials properties repository
793 from high-throughput ab initio calculations. *Computational Materials Science*
794 **58**, 227–235 (2012).
795

796 [30] Calderon, C. E. *et al.* The aflow standard for high-throughput materials science
797 calculations. *Computational Materials Science* **108**, 233–238 (2015).
798

799 [31] Landis, D. D. *et al.* The computational materials repository. *Computing in*
800 *Science & Engineering* **14**, 51–57 (2012).
801

802 [32] Draxl, C. & Scheffler, M. The nomad laboratory: from data sharing to artificial
803 intelligence. *Journal of Physics: Materials* **2**, 036001 (2019).
804

805 [33] Talirz, L. *et al.* Materials cloud, a platform for open computational science.
806 *Scientific data* **7**, 299 (2020).

807 [34] Andersen, C. W. *et al.* Optimade, an api for exchanging materials data. *Scientific*
808 *data* **8**, 217 (2021).
809

810 [35] Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation
811 made simple. *Physical review letters* **77**, 3865 (1996).
812

813 [36] Blöchl, P. E. Projector augmented-wave method. *Physical review B* **50**, 17953
814 (1994).
815

816 [37] Kresse, G. & Furthmüller, J. Efficient iterative schemes for ab initio total-energy
817 calculations using a plane-wave basis set. *Physical review B* **54**, 11169 (1996).
818

819 [38] Kresse, G. & Joubert, D. From ultrasoft pseudopotentials to the projector
820 augmented-wave method. *Physical review b* **59**, 1758 (1999).
821

822 [39] Wang, L., Maxisch, T. & Ceder, G. Oxidation energies of transition metal
823 oxides within the gga+ u framework. *Physical Review B—Condensed Matter and*
824 *Materials Physics* **73**, 195107 (2006).

825 [40] Jain, A. *et al.* Formation enthalpies by mixing gga and gga+ u calculations.
826 *Physical Review B—Condensed Matter and Materials Physics* **84**, 045115 (2011).
827
828

[41]	Stevanović, V., Lany, S., Zhang, X. & Zunger, A. Correcting density functional theory for accurate predictions of compound enthalpies of formation: Fitted elemental-phase reference energies. <i>Physical Review B—Condensed Matter and Materials Physics</i> 85 , 115104 (2012).	829 830 831 832 833
[42]	Hautier, G., Ong, S. P., Jain, A., Moore, C. J. & Ceder, G. Accuracy of density functional theory in predicting formation energies of ternary oxides from binary oxides and its implication on phase stability. <i>Physical Review B—Condensed Matter and Materials Physics</i> 85 , 155208 (2012).	834 835 836 837 838
[43]	Lejaeghere, K. <i>et al.</i> Reproducibility in density functional theory calculations of solids. <i>Science</i> 351 , aad3000 (2016).	839 840 841
[44]	Hegde, V. I. <i>et al.</i> Quantifying uncertainty in high-throughput density functional theory: A comparison of aflow, materials project, and oqmd. <i>Physical Review Materials</i> 7 , 053805 (2023).	842 843 844
[45]	Aykol, M. <i>et al.</i> Network analysis of synthesizable materials discovery. <i>Nature communications</i> 10 , 2018 (2019).	845 846 847
[46]	Bartel, C. J. <i>et al.</i> A critical examination of compound stability predictions from machine-learned formation energies. <i>npj computational materials</i> 6 , 97 (2020).	848 849 850
[47]	Ong, S. P. <i>et al.</i> Python materials genomics (pymatgen): A robust, open-source python library for materials analysis. <i>Computational Materials Science</i> 68 , 314–319 (2013).	851 852 853 854
[48]	Deng, B. <i>et al.</i> Chgnet as a pretrained universal neural network potential for charge-informed atomistic modelling. <i>Nature Machine Intelligence</i> 5 , 1031–1041 (2023).	855 856 857 858
[49]	Deng, B. <i>et al.</i> Systematic softening in universal machine learning interatomic potentials. <i>npj Computational Materials</i> 11 , 9 (2025).	859 860 861
[50]	Kuner, M. C., Kaplan, A. D., Persson, K. A., Asta, M. & Chrzan, D. C. Mp-aloe: an r2scan dataset for universal machine learning interatomic potentials. <i>npj Computational Materials</i> 11 , 352 (2025).	862 863 864
[51]	Lysogorskiy, Y., Bochkarev, A. & Drautz, R. Graph atomic cluster expansion for foundational machine learning interatomic potentials. <i>npj Computational Materials</i> (2026).	865 866 867 868
[52]	Kim, J. <i>et al.</i> Optimizing cross-domain transfer for universal machine learning interatomic potentials. <i>Nature Communications</i> (2026).	869 870 871 872 873 874